

Modeling Rate of Change and Sensitivity

Group Activity: The Sliding Ladder

Group Size: 2–3 Students

Time: 20 Minutes

Group Members: _____

Instructions: Discuss each question with your group before writing your response. For every rate you calculate, think about its **sign**, **units**, and **meaning in the physical situation**.

The Situation

A 10-ft ladder is leaning against a vertical wall. The bottom of the ladder is slowly pulled away from the wall, causing the top of the ladder to slide downward.

Let

x = distance of the bottom of the ladder from the wall,

and

y = height of the top of the ladder above the ground.

Task 1: Build the Model

- a. Draw and label a diagram of the physical situation. Identify x , y , and the length of the ladder.

- b. Which quantities are changing as the ladder moves?

- c. Which quantity remains constant?

- d. Construct an equation that relates x and y and remains true throughout the motion of the ladder.

- e. Why does this relationship remain true even though both x and y are changing?

Task 2: Predict Before Calculating

Suppose the bottom of the ladder is moving away from the wall.

- a. Is x increasing or decreasing?
- b. Is y increasing or decreasing?
- c. Without doing any calculations, predict the sign of

$$\frac{dy}{dx}.$$

Should it be positive, negative, or zero? Explain your reasoning using the physical situation.

Task 3: How Sensitive is y to x ?

Your equation from Task 1 describes a relationship between x and y .

- a. Differentiate the relationship implicitly with respect to x . Remember that y is also changing with x .

- b. Rearrange your result to find

$$\frac{dy}{dx}.$$

- c. Suppose the bottom of the ladder is 6 ft from the wall. Find the height y of the top of the ladder.

- d. At this position, calculate

$$\frac{dy}{dx}.$$

- e. What does the **sign** of your answer tell you?

- f. Interpret the **magnitude** of $\frac{dy}{dx}$ in the physical situation. Complete the statement:

“Near this position, if the bottom of the ladder moves a small distance away from the wall, then ...”

Task 4: Bring Time into the Model

Suppose the bottom of the ladder is being pulled away from the wall at a rate of

$$\frac{dx}{dt} = 2 \text{ ft/s}.$$

- a. Before calculating, predict the sign of

$$\frac{dy}{dt}.$$

Explain your prediction.

- b. Differentiate the original relationship between x and y with respect to **time**.

- c. When $x = 6$ ft, determine

$$\frac{dy}{dt}.$$

- d. Interpret your answer, including its sign and units, in the context of the ladder.

Task 5: Make the Connection

At $x = 6$ ft, your group has found three quantities:

$$\frac{dy}{dx}, \quad \frac{dx}{dt}, \quad \frac{dy}{dt}.$$

a. Find a mathematical relationship connecting these three quantities.

b. Complete the following modeling statement:

Sensitivity of y to $x \times$ Rate at which x changes =

c. Explain this relationship in your own words without using mathematical symbols.

Task 6: What Happens if We Change the Position?

Keep the pulling rate fixed at

$$\frac{dx}{dt} = 2 \text{ ft/s.}$$

Your instructor will assign your group one of the following positions:

$$x = 2, \quad x = 6, \quad x = 8, \quad x = 9 \text{ ft.}$$

For your assigned value of x :

a. Find y .

b. Find the sensitivity

$$\frac{dy}{dx}.$$

- c. Find the related rate

$$\frac{dy}{dt}.$$

- d. Record your group's results in the table.

x (ft)	y (ft)	$\frac{dy}{dx}$	$\frac{dy}{dt}$ (ft/s)

Final Group Challenge

After comparing your results with the other groups, discuss the following:

- a. At which position is the height of the ladder **most sensitive** to a change in the position of its bottom? How can you tell?

- b. As the ladder becomes more horizontal, what happens to the magnitude of

$$\frac{dy}{dx}?$$

- c. Explain physically why the same small horizontal movement of the bottom can produce different vertical movements of the top at different positions.

- d. In this activity, what is the difference between

$$\frac{dy}{dx} \quad \text{and} \quad \frac{dy}{dt}?$$

Explain what information each quantity provides about the system.

Think About the Model

When quantities in a physical system are connected, changing one quantity may force another quantity to change. A derivative can help us describe not only *whether* something changes, but also *how sensitive* one quantity is to another.